

Detecting Provider Billing Anomalies in Medicaid Claims: A Multi-Signal Peer Deviation and Autoencoder Approach [DRAFT]

ABSTRACT

Medicaid program integrity faces a persistent detection problem: fraudulent billing patterns are heterogeneous, evolve faster than rule-based detection systems, and typically surface only after years of financial exposure. This paper describes a two-stage provider-level fraud detection pipeline built on the CMS Medicaid Provider Claims Dataset, covering 162 million claim-month aggregates from 1.42 million individual providers across 2018–2024. The first stage computes peer-normalized robust z-scores across three independent fraud-relevant signals—unit price (paid per claim), billing volume, and billing intensity (claims per beneficiary)—with peer groups defined by taxonomy, state, and procedure code. The second stage trains a feed-forward autoencoder on aggregated provider-year features to identify multivariate anomalies that single-signal thresholds miss. The two methods produce ranked candidate lists with 22% overlap at top-100, confirming that threshold-based and reconstruction-based approaches detect distinct fraud archetypes. The rule-based pipeline preferentially surfaces sustained multi-dimensional outliers in pathology and high-volume primary care, while the autoencoder uniquely identifies beneficiary-churn patterns in pediatrics and pediatric dentistry—fraud types historically underserved by volume-and-price detection logic. The paper reports \$493M in peer-anomalous billing across the top 100 high-confidence candidates and discusses methodological choices including the separation of price-per-claim from aggregate paid amount, the treatment of T-MSIS reporting lag, and the handling of structurally sparse peer groups that materially affect fraud detection output. Investigation into these individual providers is necessary for validation of concerns.

Keywords: Medicaid fraud, T-MSIS, peer group analysis, autoencoder, anomaly detection, claims analytics, program integrity

1. Introduction and Research Background

1.1 The Scale and Structure of Medicaid Fraud

Medicaid is the largest public health insurance program in the United States, covering more than 80 million low-income adults, children, people with disabilities, and nursing home residents, with combined federal-state expenditures exceeding \$800 billion annually (MACPAC, 2024). The Government Accountability Office has classified Medicaid as a high-risk federal program since 2003, with estimated improper payments totaling approximately \$31.1 billion in FY 2024—roughly 5.1% of program expenditures (U.S. Government Accountability Office [GAO], 2025a). GAO has noted that the reported improper payment rate likely understates the true scale of payment integrity problems, because current Centers for Medicare & Medicaid Services methodology does not review payments from managed care plans to providers, despite managed care comprising approximately three-quarters of Medicaid spending (GAO, 2025b). While improper payments encompass administrative errors, eligibility mistakes, and documentation deficiencies in addition to fraud, intentional fraud drives substantial enforcement activity: in fiscal year 2024, Medicaid Fraud Control Units across all 50 states, the District of Columbia, Puerto Rico, and the U.S. Virgin Islands secured 1,151 convictions and recovered \$1.4 billion in criminal and civil judgments, with personal care service attendants accounting for 36 percent of fraud convictions (Office of Inspector General [OIG], 2025).

Fraud detection in Medicaid faces structural difficulties that distinguish it from private insurance fraud analytics. First, Medicaid operates as 56 separately administered programs (50 states, DC, and five territories), each with distinct provider enrollment procedures, fee schedules, and program integrity capacities (Medicaid and CHIP Payment and Access Commission [MACPAC], 2024b). Second, Medicaid populations include beneficiaries with limited ability to verify services rendered in their names—notably children and individuals with cognitive disabilities—which creates opportunities for phantom billing and beneficiary-churn fraud patterns that are difficult to detect through beneficiary self-report (Office of Inspector General [OIG], 2025). Third, enforcement action often lags fraud commission by years; FY 2024 MFCU case data shows criminal and civil Medicaid fraud prosecutions routinely involve multi-year investigation periods before conviction, reflecting the complexity of distinguishing fraudulent billing patterns from legitimate practice variation (OIG, 2025).

The detection problem has taken on new policy urgency. The One Big Beautiful Bill Act (P.L. 119-21), signed into law on July 4, 2025, will reduce federal Medicaid and CHIP spending by approximately \$1.02 trillion over the 2025-2034 period, with the Congressional Budget Office projecting that more than 10 million people will lose coverage by 2034 (Congressional Budget Office [CBO], 2025; Crowell & Moring LLP, 2025). Proponents have framed the cuts as addressing "waste, fraud, and abuse" in the program, though independent analysis finds that only seven of the act's 24 Medicaid provisions target fraud or waste directly, accounting for approximately 3.2 percent of the total spending reduction (Park & Solomon, 2025). Regardless of one's view on the legislation, the contraction of Medicaid's financial envelope sharpens the case for accurate, methodologically rigorous fraud detection: every dollar of genuine improper payment now displaces a dollar of legitimate care from a smaller pool, and every false accusation of fraud risks undermining legitimate providers in an already strained system. Notably, existing anti-fraud infrastructure offers substantial return on investment—\$3.46 in recoveries for every \$1 spent on Medicaid Fraud Control Units in FY 2024 (Office of Inspector General [OIG], 2025)—yet OBBBA did not expand funding for these programs.

1.2 The Limits of Rule-Based Fraud Detection

Traditional Medicaid fraud detection has relied heavily on rule-based logic: threshold rules on billing volume, prepayment edits, National Correct Coding Initiative (NCCI) edits for procedural unbundling, and cross-reference rules such as dead-beneficiary or excluded-provider checks (Office of Inspector General [OIG], 2017; U.S. Government Accountability Office [GAO], 2017). These systems catch gross errors and repeat offenders but have known weaknesses against three fraud archetypes: (1) novel schemes that predate any specific rule, (2) fraud patterns where no single metric crosses an alarming threshold but the combination is unusual, and (3) fraud distributed thinly across many providers rather than concentrated in individual outliers (Bauder & Khoshgoftaar, 2017; Johnson & Khoshgoftaar, 2019; Herland et al., 2018).

Academic and applied research has increasingly shifted toward statistical and machine learning methods for provider-level fraud detection. Peer-group-based z-score methods, unsupervised clustering, isolation forests, and autoencoders have all been deployed in healthcare claims analytics (Bauder et al., 2023; Mayaki & Riveill, 2022; Naidoo & Marivate, 2020). Recent reviews of unsupervised anomaly detection methods in healthcare insurance fraud confirm that isolation forests, Bayesian hierarchical models, and deep autoencoders consistently outperform traditional rule-based approaches, though gaps remain in interpretability, transfer learning, and real-time detection (Settipalli & Mandava, 2024). However, most published research operates on limited-scope datasets—individual state Medicaid extracts, Medicare Part B data, or commercial claims—rather than the full T-MSIS corpus.

1.3 Research Contribution and Scope

This paper presents a provider-level fraud detection pipeline operating on the full CMS Medicaid Claims Dataset over a 2018–2024 window. The methodological contributions are threefold:

First, the pipeline separates three independent fraud-relevant signals—unit price, volume, and intensity—that are typically conflated in aggregate-dollar metrics. This separation is empirically consequential: naïve ranking on total paid amount confounds legitimate high-volume providers (reference laboratories, group practices) with unit-price fraudsters, producing biased top candidate lists. The paper demonstrates this effect quantitatively.

Second, the pipeline combines threshold-based rule detection with a multivariate autoencoder trained on peer-normalized deviation features. The two methods are compared head-to-head, and the paper examines which fraud archetypes each method preferentially identifies.

Third, the paper documents methodological decisions like handling of T-MSIS reporting lag, treatment of generic taxonomy codes that fail peer grouping, construction of confidence tiers for statistical reliability—that are typically underspecified in healthcare fraud detection literature but materially affect output quality.

The paper does not claim to have identified specific fraudulent providers. Statistical anomaly is not evidence of fraud; it is grounds for investigation. All reported findings are statistical patterns that warrant human review by appropriate program integrity authorities.

2. Research Methods

2.1 Peer Definition

The pipeline defines peer groups by the triplet (taxonomy_group, state, HCPCS_code), where taxonomy_group is the first four characters of the provider's NUCC Healthcare Provider Taxonomy Code. This definition captures the three primary dimensions of billing context: specialty (taxonomy), market (state), and service (procedure code). Alternative peer definitions—including specialty-only, state-only, and Census-division-level—were considered but rejected: specialty-only ignores regional reimbursement and practice-pattern variation; state-only ignores specialty mix; and Census-division peer groups were reserved as a fallback for sparse cells rather than a primary definition. Peer group analysis as a fraud detection paradigm dates to Bolton and Hand (2001), who formalized the use of local anomaly detection against dynamically defined peer populations in longitudinal data; subsequent work has adapted this framework to healthcare claims specifically (Herland et al., 2018; Settipalli & Waghmare, 2021). This pipeline's approach differs from dynamic peer group analysis (Bolton & Hand, 2001) in using a priori peer definitions based on NUCC taxonomy codes rather than learned behavioral clusters. This choice prioritizes interpretability and regulatory defensibility by creating peer groups that are statable and auditable over the potentially sharper detection sensitivity of behavioral clustering.

Peer groups are computed at two temporal scopes: within-year (each of seven calendar years separately) and cross-year (pooled across 2018–2024). The within-year scope captures year-specific peer norms and detects year-over-year behavior shifts; the cross-year scope provides a stable long-term baseline against which individual years can be compared. Anomaly signals are computed at both scopes.

2.2 Three Independent Anomaly Signals

Three billing metrics are computed at the (provider, HCPCS, month) grain:

Unit price (PPC): $\text{paid_per_claim} = \text{TOTAL_PAID} / \text{TOTAL_CLAIMS}$. Detects fraud patterns including upcoding, fee inflation, unbundling, and modifier abuse—the same patterns that CMS National Correct Coding Initiative (NCCI) procedure-to-procedure and medically unlikely edits are designed to prevent at the claim level (Centers for Medicare & Medicaid Services [CMS], 2022). A 2022 review of healthcare fraud data mining methods identified upcoding, unbundling, and fragmentation of procedures as among the most commonly studied fraud types in the machine learning literature (Thornton et al., 2022).

Volume: `TOTAL_CLAIMS`. Detects phantom billing, visit-stuffing, impossible-hours schemes, and bulk-code fraud—fraud archetypes that have been formally recognized in the healthcare fraud taxonomy literature (Thornton et al., 2022) and operationalized in enforcement practice. The "impossible day" or "high-day billing" scheme—where a provider's claimed patient-contact hours exceed the 24-hour clock, sometimes by substantial margins—has produced criminal convictions in cases where simple volume analysis on Medicare claims data revealed billing patterns that were physically impossible (May, 2025). Volume-based peer deviation captures this archetype directly: a provider billing at many times the peer median for a given (taxonomy, state, HCPCS) triplet is either a high-legitimacy outlier meriting further context or a fraud case meriting investigation.

Intensity: `claims_per_beneficiary = TOTAL_CLAIMS /`

`TOTAL_UNIQUE_BENEFICIARIES`. Detects small-panel exploitation, beneficiary-churn schemes, and kickback-adjacent patterns where a small set of beneficiaries is billed repeatedly. This fraud pattern has been documented in pediatric Medicaid contexts: the 2018 False Claims Act settlement against Benevis LLC and more than 130 of its affiliated Kool Smiles dental clinics across 17 states, for example, alleged that the company systematically billed Medicaid for medically unnecessary pulpotomies, extractions, and stainless steel crowns on children, with dentists incentivized through production-based bonuses tied to revenue per patient (U.S. Department of Justice [DOJ], 2018). Intensity-based detection is methodologically well-suited to this archetype because the fraud manifests not through abnormal unit pricing or total volume, but through abnormally high claims-per-beneficiary ratios within an otherwise ordinary-looking practice.

These three metrics are structurally independent: any one can vary without the others. The aggregate metric `TOTAL_PAID`, by construction equal to `TOTAL_CLAIMS × paid_per_claim`, contains no information not already in the two underlying components and is therefore excluded from the final signal set. This methodological point is not universally observed in fraud detection literature; ranking by aggregate paid amount is common but confounds price and volume.

2.3 Robust Z-Score Deviation

For each row, a robust z-score is computed against the relevant peer group. Robust z-score statistics are used rather than mean/standard deviation because Medicaid billing distributions are heavy-tailed; parametric z-scores computed from the mean and standard deviation are dominated by a small number of extreme values, producing a masking effect in which large outliers inflate the scale estimate and cause other genuine outliers to appear normal (Rousseeuw & Hubert, 2011). The pipeline uses the median and interquartile range (IQR) as robust substitutes for the mean and standard deviation; IQR has a 25% breakdown point, meaning the estimate remains stable even when up to a quarter of the data are contaminated, a property well-suited to fraud detection contexts where a non-trivial fraction of the data may itself be anomalous (Huber, 1981; Rousseeuw & Leroy, 1987). The pipeline uses IQR rather than MAD because percentile-based

statistics are efficient to compute at scale in BigQuery using APPROX_QUANTILES, and IQR's 25% breakdown point is sufficient for the expected contamination rate in Medicaid billing data where fraudulent providers constitute a small minority of the provider population.

Peer statistics are computed using BigQuery's APPROX_QUANTILES function, which provides approximate quantiles at near-linear cost; approximation error is negligible at the peer group sizes relevant to this analysis (median peer group size > 100 providers for common HCPCS codes).

2.4 Confidence Tiers

Peer groups with fewer than five providers are excluded from scoring entirely, as the median and IQR are not statistically meaningful. Peer groups are tagged with confidence tiers based on provider count: **high** (≥30 providers), **medium** (15–29), **low** (5–14), **exclude** (<5). Only high- and medium-confidence peer groups are used in downstream anomaly flagging.

2.5 Provider-Level Rollup

For each servicing NPI, deviation signals are rolled up to the provider level using a breadth-weighted dollar formula:

$$\text{anomaly_score} = N_{\text{anomalous codes}} \times \ln(1 + D_{\text{anomalous paid}})$$

where $N_{\text{anomalous codes}}$ is the count of distinct HCPCS codes on which the provider exceeded the deviation threshold (dev > 10 IQRs), and $D_{\text{anomalous paid}}$ is the sum of paid dollars on those flagged rows. The logarithmic dollar term prevents extreme-dollar providers from dominating breadth-based ranking while preserving dollar-impact sensitivity. This prioritization approach is consistent with broader practice in healthcare fraud analytics, in which dollar-weighted ranking is used to direct limited investigative resources toward cases where expected recoveries justify the cost of audit (Shekhar et al., 2026; Thornton et al., 2022). Recent work on machine-learning-based Medicare fraud detection similarly aggregates per-provider anomaly signals into a single ranking for enforcement prioritization, rather than treating detection and prioritization as independent steps (Shekhar et al., 2026).

Three signal-specific rollups are computed (PPC, volume, intensity) plus one combined rollup flagging any-of-three. The separation enables fraud-archetype-specific ranking and comparison.

2.6 Autoencoder for Multivariate Anomaly Detection

The three signal-specific anomaly scores computed in Section 2.5 treat each signal independently. To capture patterns where fraud manifests through the *combination* of signals rather than any single signal in isolation, a fully-connected autoencoder is trained on provider-year feature vectors derived from the deviation table. An autoencoder learns a compressed representation of the input distribution by minimizing reconstruction error between

input and output; providers whose feature vectors the model cannot reconstruct well are, by construction, ones whose billing patterns are atypical relative to the bulk of the provider population (Hinton & Salakhutdinov, 2006; Goodfellow et al., 2016).

Features

Each provider-year is represented by a 15-dimensional feature vector:

- **PPC deviation features (4):** max, 90th percentile, mean, and threshold-exceedance count of per-HCPCS paid-per-claim deviation scores across all codes billed by the provider in that year
- **Volume deviation features (4):** same four aggregations computed on volume deviation scores
- **Intensity deviation features (4):** same four aggregations computed on intensity (claims-per-beneficiary) deviation scores
- **Scale and diversity features (3):** distinct HCPCS code count, natural log of total paid dollars, and Shannon entropy of the provider's HCPCS-level paid-dollar distribution

The scale and diversity features are included so that the autoencoder can learn context-dependent reconstruction: a high maximum PPC deviation is more suspicious for a provider billing ten codes than for one billing two hundred codes, and the entropy feature allows the model to distinguish diversified general practice from concentrated specialty billing.

Winsorization

Raw deviation features are winsorized at ± 50 IQRs prior to standardization. This bound was chosen empirically: the 99th percentile of each deviation signal sits below 30 IQRs across the scored population, so a ± 50 ceiling retains essentially all genuine anomaly signal while truncating the right tail of mathematical artifacts produced by near-zero peer IQR denominators. Without winsorization, a small number of provider-years with deviation scores in the thousands would dominate the training loss, preventing the autoencoder from learning the structure of the bulk distribution.

Architecture and training

Features are standardized via z-score normalization inside the BigQuery ML `TRANSFORM` clause, which embeds the normalization statistics into the trained model so that inference-time scoring applies identical transformations. The network architecture is a symmetric ($15 \rightarrow 10 \rightarrow 6 \rightarrow 10 \rightarrow 15$) fully-connected autoencoder with ReLU activations and 10% dropout regularization in the hidden layers. The model is trained for up to 20 iterations with BigQuery ML's built-in early stopping (`early_stop=TRUE`, `min_rel_progress=0.01`), optimizing mean squared reconstruction error.

Train/holdout split

Provider-year rows are partitioned into 80/20 train/holdout sets using a deterministic hash of the (NPI, year) pair rather than random sampling. The deterministic split ensures reproducibility across pipeline runs and allows the same holdout set to be tracked across model iterations.

The split is performed at the row level rather than the provider level. This choice reflects the role the holdout serves in the pipeline: the autoencoder is not used for predictive generalization to unseen providers but for reconstruction-error scoring of every provider-year in the dataset. The holdout split exists narrowly to verify that the trained model has not catastrophically overfit—holdout reconstruction error should track training reconstruction error closely. A provider-level split would be the appropriate design if the goal were predictive generalization to new providers, but would not change the scoring procedure, which operates on all provider-years regardless of split membership. Observed holdout MSE (0.4170) was within 1.2% of training MSE (0.4122), indicating stable generalization and no evidence of memorization.

Scoring and provider-level rollup

At inference, each provider-year feature vector is passed through the trained autoencoder, and mean squared error between input and reconstruction is computed as the per-row anomaly score. Because the pipeline's unit of investigative action is the provider (not the provider-year), per-year scores are rolled up to the provider level by taking the maximum reconstruction error across the provider's year-rows.

The maximum was chosen over alternatives such as the mean or sum. Ranking by mean dilutes the signal for providers whose fraud is concentrated in a single year, which is the empirically dominant pattern in the MFCU case histories reviewed in Section 1. Ranking by sum introduces a bias toward providers with more years of data, penalizing newer providers and rewarding longer-tenured ones regardless of billing behavior. Ranking by maximum is sensitive to single-year data-quality artifacts, but this risk is mitigated by the peer confidence tiering (Section 2.4) and by the winsorization applied before training. The tradeoff—accepting some sensitivity to single-year noise in exchange for preserving sustained-anomaly signal—is made explicit here so that future work can benchmark alternative rollup strategies.

Relationship to the rule-based score

The autoencoder score and the breadth-weighted dollar score are treated as complementary rather than redundant. The rule-based score captures providers who are anomalous on multiple signals simultaneously, with dollar impact. The autoencoder score captures providers whose *combination* of signals is unusual even when no single signal crosses a simple threshold. Results section presents evidence that these two scores surface substantially different provider populations: of the top 100 providers ranked by each score, only 22 appear on both lists, with 78 unique to each. This divergence is treated not as a failure of either method but as evidence that the two scoring approaches are sensitive to different fraud archetypes.

3. Materials and Data Sources

3.1 Center for Medicare & Medicaid Services Provider Claims Dataset

The primary data source is a provider-level Medicaid spending dataset released by the Centers for Medicare & Medicaid Services on February 14, 2026 at opendata.hhs.gov, drawn from Transformed Medicaid Statistical Information System (T-MSIS) submissions and aggregated by billing or servicing provider, procedure code, and month (CMS, 2026). The dataset covered fee-for-service, managed care, and CHIP claims from January 2018 through December 2024 and was framed by CMS as enabling public identification of unusual billing patterns.

The `provider_claims` base table derived from this source contains 162 million rows at the (servicing_NPI, billing_NPI, HCPCS, month) grain. Each row aggregates all claims with matching dimensions and reports `TOTAL_CLAIMS` (count of claim lines), `TOTAL_UNIQUE_BENEFICIARIES` (distinct patients served), and `TOTAL_PAID` (sum of Medicaid payment amounts).

3.2 National Plan and Provider Enumeration System (NPES)

Provider-level attributes — Healthcare Provider Taxonomy Code, practice state, practice ZIP code, sole-proprietor status, and entity type — are joined from the National Plan and Provider Enumeration System (NPES), the public federal registry of NPI-identified healthcare providers maintained by CMS and made available through the NPI Registry and the monthly-refreshed NPES Downloadable File (Centers for Medicare & Medicaid Services [CMS], 2025a). NPES is FOIA-disclosable, unencrypted, and includes both active and deactivated NPIs along with their deactivation dates.

Providers are filtered to NPES entity type 1 (individual providers) rather than type 2 (organizations), ensuring that the servicing NPI corresponds to a natural person and that subsequent peer-group comparisons are across comparable practice units. Deactivated NPIs are excluded from the scored population unless reactivated within the study period; deactivation is retained as a joinable attribute rather than used as a filter, to preserve the ability to analyze billing activity immediately preceding deactivation as a potential signal for future work.

3.3 Filtering and Cleaning

Three data quality corrections were applied to the raw extract:

State field cleaning: The NPES state field, entered as free text during NPI registration, contained 166 distinct values including full state names, foreign addresses, military codes, and typos. The field was normalized to the 52 valid US postal codes (50 states, DC, PR); approximately 100,000 rows (0.06% of total) were dropped.

Reporting lag trimming: State submissions to T-MSIS accumulate over time as claims adjudication completes, and CMS considers a service month "fully mature" only after approximately twelve months of runout for each service month (Centers for Medicare &

Medicaid Services [CMS], 2025a; Research Data Assistance Center [ResDAC], 2025). Because the upstream provider-spending dataset was released in February 2026 and drew on submissions through mid-2025, the most recent months of 2024 data were structurally incomplete at ingestion time.

Analysis of monthly row counts against a trailing-six-month baseline revealed a sharp fall-off in the final two months of the study window: November 2024 rows were at approximately 75% of baseline and December 2024 rows at approximately 27% of baseline, indicating substantial undercount consistent with typical T-MSIS reporting lag patterns. Service months after October 2024 were excluded from all downstream analysis.

Generic taxonomy flagging: Two NUCC taxonomy codes—174400000X ("Specialist") and 390200000X ("Student, Health Care")—are structurally incoherent for peer comparison because they group providers performing wildly different services. These were retained in the dataset but flagged; their downstream anomaly scores are treated with reduced confidence.

3.4 Computational Environment

All analytical queries were executed in Google BigQuery, using BigQuery ML for autoencoder training. The pipeline produces intermediate tables at each stage (peer statistics, deviation scores, provider rollups, autoencoder features, reconstruction errors) to support inspection and iteration without recomputation.

4. Results

4.1 Pipeline Coverage

After filtering, the analytical dataset contained 162 million claim-month rows representing 1.42 million unique servicing providers across 7,984 distinct HCPCS codes and 52 US jurisdictions. Seven calendar years (2018–2024, with 2024 through October only) were included.

Peer groups at the (taxonomy, state, HCPCS) level showed good coverage for mainstream specialties: 79–82% of rows belonged to peer groups meeting the high-confidence threshold (≥ 30 providers). Cross-year pooled peer groups showed even higher coverage, with approximately 88% of rows qualifying as high-confidence. Exclusion rates were consistent across all seven years, indicating no year-specific coverage collapse.

4.2 Deviation Score Distribution

Price-per-claim deviations, computed cross-year on high and medium confidence peer groups, showed the expected distribution for a robust z-score metric: median near zero (by mathematical construction), 90th percentile at 1.76, 99th percentile at 7.88, and 99.9th percentile at 49.7. A pathological tail extending to 7,260+ at the 99.99th percentile reflects math

artifacts in peer groups with near-zero IQRs; these were handled by winsorization in the autoencoder feature set and by directional threshold filtering in the rule-based rollup.

Similar distributions were observed for volume and intensity deviations. A threshold of dev > 10 was adopted, representing approximately the 99.4th percentile of the deviation distribution, aligned with the natural distributional break between "unusual" and "math artifact" regimes.

4.3 Rule-Based Candidate List

Applying the three-signal rule (provider flagged if the cross-year deviation on any of PPC, volume, or intensity exceeded 10 for at least one code), the pipeline identified:

- **198,608 providers (14.0%)** flagged on at least one signal
- **85,046 providers (6.0%)** flagged on PPC
- **30,260 providers (2.1%)** flagged on volume
- **125,428 providers (8.8%)** flagged on intensity
- **4,553 providers (0.32%)** flagged on all three signals

The all-three-signal intersection represents a 30× enrichment over the product of individual rates, indicating that anomalies cluster across signals rather than distributing independently, which is consistent with the hypothesis that multi-dimensional anomaly is a coherent fraud signature.

Among the top 100 providers by combined anomaly score (restricted to the all-three-signal population), the taxonomy distribution was:

Taxonomy	Count	Specialty
2080	49	Pediatrics
207Z	22	Pathology
207Q	10	Family Medicine
207R	9	Internal Medicine
207P	7	Emergency Medicine
2085	2	Radiology
207V	1	OB/GYN

Total peer-anomalous billing across these 100 providers was **\$493 million** over the 7-year window. Per-provider averages ranged from \$850K (Radiology) to \$7.4M (Family Medicine). State distribution tracked total state volume, with California (31), Florida (21), New York (10), and Texas (9) accounting for 71% of the top 100.

4.4 Autoencoder Candidate List

The autoencoder trained successfully with minimal overfitting: train-set average reconstruction MSE was 0.4122 versus holdout-set average 0.4170, a 1.2% difference. The p50/p99 ratio within the holdout was 22×, indicating strong separation between typical and anomalous reconstruction.

The top 100 providers by worst-year reconstruction error showed a markedly different specialty distribution than the rule-based list. Pediatrics remained dominant, but pediatric dentistry (NUCC 1223) which was absent from the rule-based top 100 appeared repeatedly among the top 20 autoencoder candidates.

Feature inspection revealed the reason: the flagged dentists exhibited extreme intensity deviations (often hitting the winsorization cap of 50) combined with moderate-to-low price and volume deviations. Because the rule-based system required non-zero scores on all three signals, a dentist with normal price and volume patterns but extreme intensity was automatically disqualified from the top list, even though extreme intensity alone is a documented Medicaid pediatric dental fraud signature. The Office of Inspector General's own analytical work on pediatric Medicaid dental fraud uses precisely this signal: OIG's 2014 review of California pediatric dental billing flagged providers who performed extractions on 44% or more of the children they served — compared to a state average of 12% — with the most extreme flagged provider performing extractions on 98% of served children (OIG, 2014). This pattern has been documented in two of the largest pediatric dental Medicaid fraud settlements in U.S. history: FORBA Holdings/Small Smiles (U.S. Department of Justice [DOJ], 2010) and Benevis/Kool Smiles (DOJ, 2018), both of which were alleged to have used production-based bonus structures to incentivize dentists to perform pulpotomies, extractions, and stainless steel crowns on Medicaid-insured children at rates far exceeding clinical necessity.

4.5 Comparison of the Two Methods

Head-to-head comparison of the two top-100 lists produced:

- **22 providers in both lists**
- **78 in rule-based only**
- **78 in autoencoder only**

The 22-provider overlap represents approximately 3,000× random expected overlap, confirming that both methods detect real signals. The 78% disagreement, however, indicates that the methods identify substantially different fraud archetypes.

Rule-based-only providers were predominantly concentrated in pathology and sustained high-volume practices in large states where multi-dimensional extreme deviations co-occur. The rule-based system's strength is identifying long-running, high-dollar, multi-signal outliers.

Autoencoder-only providers included pediatric dentists, narrow-specialty pediatricians, and providers with unusual feature-combination shapes. These are patterns where no single

threshold is crossed dramatically but the overall billing profile is structurally unusual relative to the learned population distribution.

The complementary nature of the two methods suggests that they should be used in combination rather than as alternatives. The 22 "both-list" providers represent the highest-confidence candidate set; the 78 rule-only and 78 autoencoder-only providers each capture fraud patterns the other method misses.

4.6 Case Study: A Flagged Pediatric Billing Pattern

To illustrate the qualitative character of flagged providers, one autoencoder-top-ranked provider was examined in detail. The provider, registered in NPPES as a pediatrician in a major metropolitan area, showed seven years of sustained elevated reconstruction error, with the highest single-year score in the top 0.0001% of the population.

Monthly claim inspection revealed two distinct billing NPIs using the physician as the servicing provider: one associated with pediatric well-child and established-patient E/M codes, another associated with urgent-care-style codes (99051 after-hours, 99203/99204 new-patient E/M, 12001 wound repair, 73130 extremity X-ray). Combined monthly claim volume reached approximately 1,900 claims per month in April 2018, distributed across both billing entities.

This pattern is consistent with, but not conclusive evidence of, a "sham medical director" or improper "incident-to" billing arrangement, in which a credentialed physician's NPI is used as the servicing provider of record for services rendered by non-physician staff at multiple sites. OIG has repeatedly raised concerns that incident-to billing, by allowing multiple individuals to deliver services under a single identification number, creates a lack of transparency that hinders CMS's ability to determine whether a provider is billing for a reasonable volume of services, but CMS has declined to adopt the addition of a service-code modifier that would make incident-to billing visible at the claim level (Bipartisan Policy Center, 2026). In the absence of such a modifier, claims data alone cannot distinguish between legitimate incident-to billing and a sham-medical-director arrangement.

5. Discussion

5.1 Methodological Findings

Three methodological findings from this work have implications for healthcare fraud detection practice.

Price and volume must be separated. Aggregate paid amount, while intuitive as a fraud signal, mathematically conflates price and volume anomalies. Ranking on aggregate paid amount produces candidate lists dominated by structurally high-volume providers (reference laboratories, group practices, hospital-based specialists) regardless of whether their unit prices are anomalous. The pipeline's initial output, before this methodological correction, identified five

Florida/Illinois pathologists as its top candidates—a concentration that appeared initially to support documented Florida pathology fraud patterns but, on inspection of per-unit prices, reflected legitimate reference lab volume rather than unit-price fraud. Correcting the methodology dropped three of the five pathologists from the top 5, replacing them with providers exhibiting genuine price anomalies.

Confidence tiers should be primary, not afterthought. Peer comparison only works when peer groups are statistically adequate. The pipeline's confidence tier system (high/medium/low/exclude based on peer group size) acts as a principled constraint on where anomaly detection is applied. Attempting to score rare-HCPCS-in-small-state cells produces noise that, at the population level, can dominate the top candidate list. This finding generalizes: any fraud detection system using peer comparison should explicitly model its own coverage, not assume uniform applicability.

Multivariate and univariate detection are complementary, not substitutes. The 22% overlap between rule-based and autoencoder top-100 lists, combined with coherent differences in the fraud archetypes each method preferentially identifies, suggests that production fraud detection should employ both approaches simultaneously rather than select between them. The rule-based system excels at transparent, auditable flagging of extreme multi-signal outliers; the autoencoder excels at identifying unusual feature combinations that evade threshold logic.

5.2 Fraud Archetype Findings

The pipeline surfaced two distinct fraud archetypes with different statistical signatures:

Sustained multi-dimensional outliers (rule-based). Providers who bill anomalously on price, volume, and intensity simultaneously, maintaining these patterns across multiple years. The flagged population is concentrated in pathology (where reference lab concentration creates overlapping high-volume and price-elevated signals) and in high-volume primary care practices. Prior literature documents substantial enforcement activity against pathology reference labs, particularly in Florida, exemplified by Operation Double Helix — a September 2019 coordinated federal takedown that charged 35 defendants with \$2.1 billion in fraudulent Medicare genetic testing billing, with defendant concentrations in Florida, Georgia, Louisiana, and Texas (DOJ, 2019; OIG, 2019).

Beneficiary-churn intensity fraud (autoencoder). Providers who bill standard-priced services to small panels of beneficiaries repeatedly, without triggering price or volume thresholds. The flagged population is concentrated in pediatrics and pediatric dentistry. This pattern is consistent with documented Medicaid dental fraud cases involving repeated unnecessary procedures on pediatric beneficiaries (DOJ, 2010; DOJ, 2018; OIG, 2014a).

The autoencoder's ability to surface the beneficiary-churn archetype despite the rule-based system's inability to rank it highly is a practical contribution of multivariate methods. Beneficiary-churn fraud is particularly damaging because it typically targets children who cannot self-report over-billing, and because individual claims appear legitimate in isolation — detection

requires either pattern analysis across many claims or external verification such as patient interviews or chart review.

5.3 Limitations

The work has substantive limitations that should be explicit:

No ground truth validation. The available suspensions list did not overlap the flagged provider population, precluding quantitative assessment of precision and recall. Future work should incorporate the OIG LEIE or state MFCU conviction records as validation sources.

Structural bias toward large states and common specialties. Peer group confidence requirements mean the pipeline performs best in high-provider-density combinations. Rare-specialty rural providers, providers in small territories, and providers in generic taxonomy categories are systematically under-scored. Fraud in these populations is not undetectable in principle but requires a different methodology.

Reporting lag asymmetry. The exclusion of November and December 2024 prevents pipeline results from reflecting the most recent billing patterns. In production, a rolling lag buffer would be required.

Provider-level attribution is imperfect. Services billed under one physician's NPI but rendered by others (legitimate incident-to billing, illegal sham arrangements, or credential-sharing fraud) produce statistical signals attributed to the credentialed provider. Distinguishing these cases requires information outside claims data.

Servicing NPI focus misses organizational fraud. The pipeline's focus on individual providers (NPPEs entity type 1) excludes fraud patterns operating at the organizational level like shell clinic arrangements, rapidly-rotating billing entities, and multi-site coordinated schemes. Billing-NPI relational features are identified as future work.

5.4 Future Work

Three extensions would materially strengthen the pipeline:

Billing NPI relational features. Per-servicing-provider features such as billing NPI diversity, billing NPI stability over time, and co-occurrence with other servicing providers would capture organizational fraud signatures invisible to current features.

Temporal anomaly modeling. The current autoencoder is cross-sectional at provider-year grain. A recurrent or transformer-based model over provider-month sequences could capture trajectory-based fraud signatures (onset patterns, escalation curves, post-enforcement behavior) more effectively.

Ground truth validation at scale. Integration with OIG LEIE, state MFCU records, and False Claims Act settlement data would enable precision/recall measurement and threshold

calibration. Without this, candidate lists are empirically meaningful anomaly measures but not validated fraud predictions.

6. Conclusion

This paper demonstrated a two-stage Medicaid fraud detection pipeline operating on seven years of CMS claims data. The rule-based peer-deviation component identifies sustained multi-dimensional outliers; the autoencoder component identifies multivariate anomalies that single-signal thresholds miss. Together, the two methods surface approximately 180 candidate providers, representing approximately \$500M in peer-anomalous billing and two distinct fraud archetypes—sustained high-volume outliers and beneficiary-churn patterns.

The methodological contribution of explicit separation of price-per-claim from volume, the confidence tier system, and the comparison framework between threshold-based and reconstruction-based methods generalizes beyond this specific pipeline. The empirical finding that autoencoder methods preferentially surface pediatric dental fraud patterns that rule-based methods miss has implications for program integrity practice.

The paper does not validate findings against ground-truth fraud records, it does not quantify precision or recall, and it does not distinguish statistical anomaly from actual fraud. These gaps are typical of unsupervised provider-level fraud detection research (Shekhar et al., 2026), and they motivate the framing of pipeline output as investigation candidates rather than fraud determinations.

REFERENCES

1. Medicaid and CHIP Payment and Access Commission. (2024). *MACStats: Medicaid and CHIP data book*. <https://www.macpac.gov/macstats/> Centers for Medicare & Medicaid Services. (2025). *Medicaid expenditure reports from MBES/CBES, fiscal year 2024 preliminary estimates*. <https://www.medicaid.gov/medicaid/financial-management/state-budget-expenditure-reporting-for-medicaid-and-chip/expenditure-reports-mbes/cbes>
2. U.S. Government Accountability Office. (2025a). *High-risk series: Heightened attention could save billions more and improve government efficiency and effectiveness* (GAO-25-107743). <https://www.gao.gov/products/gao-25-107743> U.S. Government Accountability Office. (2025b). *Medicaid managed care: Improper payment estimate* (GAO-25-107770). <https://www.gao.gov/products/gao-25-107770>
3. Medicaid and CHIP Payment and Access Commission. (2024b). *Federal requirements and state options: How states exercise flexibility under a Medicaid state plan*. <https://www.macpac.gov/publication/federal-requirements-and-state-options/>
4. Office of Inspector General, U.S. Department of Health and Human Services. (2025). *Medicaid Fraud Control Units annual report: Fiscal year 2024* (OEI-09-25-00090). <https://oig.hhs.gov/reports/all/2025/medicaid-fraud-control-units-annual-report-fiscal-year-2024/>
5. Congressional Budget Office. (2025, June 4). *Estimated budgetary effects of Public Law 119-21, the One Big Beautiful Bill Act*. <https://www.cbo.gov/>
6. Park, E. (2025, June 6). *Fraud and abuse against Medicaid: Rhetoric and reality in the One Big Beautiful Bill Act*. Georgetown University Center for Children and Families. <https://ccf.georgetown.edu/2025/06/06/fraud-and-abuse-against-medicaid-rhetoric-and-reality-in-the-one-big-beautiful-bill-act/> One Big Beautiful Bill Act, Pub. L. No. 119-21, 139 Stat. 1 (2025).
7. Office of Inspector General, U.S. Department of Health and Human Services. (2017). *The Centers for Medicare & Medicaid Services could improve performance measures associated with the Fraud Prevention System* (A-01-15-00509). <https://oig.hhs.gov/reports/all/2017/the-centers-for-medicare-medicaid-services-could-improve-performance-measures-associated-with-the-fraud-prevention-system/>
8. U.S. Government Accountability Office. (2017). *Medicare: CMS Fraud Prevention System uses claims analysis to address fraud* (GAO-17-710). <https://www.gao.gov/products/gao-17-710>
9. Bauder, R. A., & Khoshgoftaar, T. M. (2017). Medicare fraud detection using machine learning methods. *2017 16th IEEE International Conference on Machine Learning and Applications (ICMLA)*, 858–865. <https://doi.org/10.1109/ICMLA.2017.00-48>

10. Johnson, J. M., & Khoshgoftaar, T. M. (2019). Medicare fraud detection using neural networks. *Journal of Big Data*, 6(1), 63.
<https://doi.org/10.1186/s40537-019-0225-0>
11. Herland, M., Khoshgoftaar, T. M., & Bauder, R. A. (2018). Big data fraud detection using multiple Medicare data sources. *Journal of Big Data*, 5(1), 29.
<https://doi.org/10.1186/s40537-018-0138-3>
12. Bauder, R. A., Herland, M., & Khoshgoftaar, T. M. (2023). Explainable machine learning models for Medicare fraud detection. *Journal of Big Data*, 10(1), 154.
<https://doi.org/10.1186/s40537-023-00821-5>
13. Naidoo, K., & Marivate, V. (2020). Unsupervised anomaly detection of healthcare providers using generative adversarial networks. In M. Hattingh, M. Matthee, H. Smuts, I. Pappas, Y. K. Dwivedi, & M. Mäntymäki (Eds.), *Responsible design, implementation and use of information and communication technology* (pp. 419–430). Springer. https://doi.org/10.1007/978-3-030-44999-5_35
14. Settipalli, L., & Mandava, M. (2024). A review of unsupervised anomaly detection techniques for health insurance fraud. In *2024 International Conference on Sustainable Communication Networks and Applications (ICSCNA)*. IEEE.
<https://doi.org/10.1109/ICSCNA63714.2024.10729915>
15. Bolton, R. J., & Hand, D. J. (2001). *Peer group analysis: Local anomaly detection in longitudinal data* (Technical Report). Department of Mathematics, Imperial College London.
16. Settipalli, L., & Waghmare, G. R. (2021). Healthcare fraud detection using primitive sub peer group analysis. *Concurrency and Computation: Practice and Experience*, 33(23), e6275. <https://doi.org/10.1002/cpe.6275>
17. Centers for Medicare & Medicaid Services. (2022). *National Correct Coding Initiative policy manual for Medicaid services, Chapter I: General correct coding policies*.
<https://www.medicare.gov/medicaid/program-integrity/downloads/nccimanual2022-chapterone.pdf>
18. Kumaraswamy N, Markey MK, Ekin T, Barner JC, Rascati K. Healthcare Fraud Data Mining Methods: A Look Back and Look Ahead. *Perspect Health Inf Manag*. 2022 Jan 1;19(1):1i. PMID: 35440932; PMCID: PMC9013219.
19. U.S. Department of Justice. (2018, January 10). *Dental management company Benevis and its affiliated Kool Smiles dental clinics to pay \$23.9 million to settle False Claims Act allegations relating to medically unnecessary pediatric dental services* [Press release].
<https://www.justice.gov/archives/opa/pr/dental-management-company-benevis-and-its-affiliated-kool-smiles-dental-clinics-pay-239>
20. May, C. (2025, April). The impossible day: High-day billing health care fraud schemes. *Association of Certified Fraud Examiners Insights Blog*.
<https://www.acfe.com/acfe-insights-blog/blog-detail?s=impossible-day-health-care-fraud-schemes>
21. Thornton, D., van Capelleveen, G., Poel, M., van Hillegersberg, J., & Mueller, R. M. (2022). Healthcare fraud data mining methods: A look back and look ahead.

- Perspectives in Health Information Management*, 19(Spring), 1i. [Verify exact pagination on PubMed — PMC9013219]
22. Huber, P. J. (1981). *Robust statistics*. John Wiley & Sons.
 23. Rousseeuw, P. J., & Hubert, M. (2011). Robust statistics for outlier detection. *WIREs Data Mining and Knowledge Discovery*, 1(1), 73–79. <https://doi.org/10.1002/widm.2>
 24. Rousseeuw, P. J., & Leroy, A. M. (1987). *Robust regression and outlier detection*. John Wiley & Sons.
 25. Shekhar, S., Leder-Luis, J., & Akoglu, L. (2026). Can machine learning target health care fraud? Evidence from Medicare hospitalizations. *Journal of Policy Analysis and Management*, 45, e70078. <https://doi.org/10.1002/pam.70078>
 26. Fang, H., & Gong, Q. (2017). Detecting potential overbilling in Medicare reimbursement via hours worked. *American Economic Review*, 107(2), 562–591. <https://doi.org/10.1257/aer.20150916>
 27. Centers for Medicare & Medicaid Services. (2026). *Medicaid provider spending* [Dataset]. U.S. Department of Health and Human Services Open Data Portal. <https://opendata.hhs.gov/datasets/medicaid-provider-spending/>
 28. Tolbert, J., & Rudowitz, R. (2026, February 24). *What newly released Medicaid data do and don't tell us*. KFF. <https://www.kff.org/medicaid/what-newly-released-medicaid-data-do-and-dont-tell-us/>
 29. Centers for Medicare & Medicaid Services. (2025a). *NPPES data dissemination: National Provider Identifier downloadable files*. U.S. Department of Health and Human Services. <https://www.cms.gov/medicare/regulations-guidance/administrative-simplification/data-dissemination>
 30. Centers for Medicare & Medicaid Services. (2025a). *Transformed Medicaid Statistical Information System (T-MSIS) Analytic Files (TAF)*. Medicaid.gov. <https://www.medicaid.gov/medicaid/data-systems/macbis/medicaid-chip-research-files/transformed-medicaid-statistical-information-system-t-msis-analytic-files-taf>
 31. U.S. Department of Justice. (2010, January 20). *National dental management company pays \$24 million to resolve fraud allegations* [Press release]. <https://www.justice.gov/opa/pr/2010/January/10-civ-052.html>
 32. Bipartisan Policy Center. (2026, March). *Comments on CMS RFI related to Comprehensive Regulations to Uncover Suspicious Healthcare (CRUSH)*. <https://bipartisanpolicy.org/testimony-letter/rfi-related-to-comprehensive-regulations-to-uncover-suspicious-healthcare-crush/>
 33. U.S. Department of Justice. (2019, September 27). *Federal law enforcement action involving fraudulent genetic testing results in charges against 35 individuals responsible for over \$2.1 billion in losses in one of the largest health care fraud schemes ever charged* [Press release]. <https://www.justice.gov/opa/pr/federal-law-enforcement-action-involving-fraudulent-genetic-testing-results-charges-against>

34. Office of Inspector General, U.S. Department of Health and Human Services. (2019). *Nationwide genetic testing fraud takedown: Operation Double Helix*. <https://oig.hhs.gov/newsroom/media-materials/media-materials-nationwide-genetic-testing-fraud/>